

Concentrations of antibiotic residues vary between different edible muscle tissues in poultry.

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Antibiotics are used by veterinarians and producers to treat disease and improve animal production. The federal government, to ensure the safety of the food supply, establishes antibiotic residue tolerances in edible animal tissues and determines the target tissues (e.g., muscle) for residue monitoring. However, when muscle is selected as the target tissue, the federal government does not specify which type of muscle tissue is used for monitoring (e.g., breast versus thigh). If specific muscle tissues incorporate residues at higher concentrations, these tissues should be selected for residue monitoring. To evaluate this possibility in poultry, chickens were divided into four groups and at 33 days of age were dosed with enrofloxacin (Baytril), as per label directions, at either 25 ppm for 3 days, 25 ppm for 7 days, 50 ppm for 3 days, or 50 ppm for 7 days. Breast and thigh muscle tissues were collected from each bird (n = 5 birds per day per group) during the dosing and withdrawal period, and fluoroquinolone concentrations were determined. The results indicate higher overall enrofloxacin concentrations in breast versus thigh muscle for each treatment group ($P < 0.05$). These data indicate, at least for enrofloxacin, that not all muscle tissues incorporate antibiotics at the same concentrations. These results may be helpful to regulatory agencies as they determine what tissues are to be monitored to ensure that the established residue safety tolerance levels are not exceeded.